



Nursing Practice Guideline

Lidocaine (Lignocaine) Infusion for Acute Pain

Scope

Site	Service/Department/Unit	Disciplines
SCGH	Inpatients, Acute Pain Service	Medical, Nursing, Pharmacy

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Introduction

This guideline outlines the responsibilities and procedures of the individual and health service organisation in relation to ward-based intravenous or subcutaneous lidocaine infusions for acute pain management.

Lidocaine is an amino-amide local anaesthetic with analgesic, anti-hyperalgesic and anti-inflammatory effects¹. Perioperative lidocaine infusion is a useful component of multimodal analgesia in patients whom other medications are contraindicated or as an additional option for patients who are at risk of significant postoperative pain problems. It is a valuable adjunct to opioid sparing, multimodal anaesthesia.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☐
Other:	☐		

Definitions

Lidocaine (Lignocaine)	An amino-amide local anaesthetic which acts by blocking sodium channels ² on nerves, thus reducing nociceptive (pain) transmission. At low systemic doses, lidocaine blocks nociceptive neurons without interfering with normal sensory, motor or cardiac function
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1. General Information

1.1. Legislative Requirements

It is a legal requirement that all disciplines that prescribe or administer medications comply with the Medicines and Poisons Act 2014, Medicines and Poisons Regulations 2016, Pharmacy Act 2010 and Hospital Guidelines. It is a legislative requirement that a valid prescription must exist before Schedule 4 medications are administered.

Only valid prescriptions will be utilised by any nursing staff to administer medication.

Refer to NPG 4 Intravenous Therapy for Principles/Safety/Procedure.

Observe the five moments for hand hygiene and aseptic non-touch technique. Follow the risk assessment outlined SCGOPHCG Infection Prevention and Management Manual - Aseptic Technique Policy. Appropriate Personal Protective Equipment (PPE) must be worn to prevent contamination and mucosal or conjunctival splash injuries.

1.2. Indications

Intravenous or subcutaneous lidocaine may be prescribed as a component of multi-modal analgesia in the following situations

1. Major surgery where regional analgesia is contraindicated, refused or failed
2. Abdomino-pelvic surgery
3. Complex spinal surgery
4. Pre-existing chronic pain
5. Opioid-tolerant patients
6. High risk of chronic neuropathic pain (e.g. thoracic surgery, limb amputation, breast surgery)
7. Severe acute pain: neuropathic pain or suspected hyperalgesia

8. Severe headache
9. Pain from ischaemic limbs

1.3. Safety: Contraindications

Absolute:

- Patient refusal
- Known lidocaine allergy (or prilocaine allergy)
- Myasthenia gravis
- Concurrent administration of local anaesthesia via another route:
 - Lidocaine infusion may be initiated four hours after the last dose of anaesthetic via epidural or regional catheter and one hour post cessation of epidural/regional infusion (without loading bolus)
 - Cease Lidocaine infusion more than four hours before establishing a new regional analgesia block

Relative:

- Cardiac arrhythmias (, conduction block, pacemaker/implantable cardiac defibrillator)
- Unstable angina, recent ACS
- Severe congestive cardiac failure
- Acute cardiovascular instability (SBP < 100 mmHg; pulse < 50 or > 110 bpm), shock
- Severe renal disease
- Severe liver disease
- Altered conscious state
- History of seizure disorder
- Pregnancy or breast feeding
- Potential for drug interactions: opioids, ketamine, ropivacaine, sodium channel blocking agents' phenytoin, tricyclic antidepressants, beta-blockers, non-dihydropyridine calcium channel blockers (diltiazem, verapamil), amiodarone
- Uncorrected electrolyte abnormalities
- Uncorrected acidaemia

1.4. Safety: Cautions and toxicity

- Loading dose should be no more than 1.5mg/kg over 2-4 min; subsequent lidocaine infusion should be no more than 2mg/kg/hour⁵.
- Suggest caution extending infusion beyond 72 hours⁵, although continuous infusion up to 14 days without toxicity reported⁶.
- Multiple channel blockade occurs at increasing plasma concentrations and can lead to cardiac and neurotoxicity.¹.
- In the Central Nervous System (CNS) this generally results in excitatory phenomena, followed by depression (i.e. significant toxicity involves seizures and then respiratory depression and arrest). This is exacerbated in the setting of acidosis (respiratory or metabolic). Classical premonitory symptoms (circumoral tingling, metallic taste, visual disturbance, presyncope, nausea) may or may not occur.
- In the cardiac system there may be bradyarrhythmia's, negative inotropy and vasoactive effects with hypotension.
- Tachyphylaxis occurs with ongoing administration (this is one of the reasons that other local anaesthetics are preferred for regional analgesia infusions).

2. Medical Management

2.1 Pre-infusion assessment - Anaesthetist

- Complete pain assessment
- Review vital signs: BP, pulse rate, sedation score, neurological state
- Check allergies and absolute/relative contraindications
- Review ECG, review urea and electrolytes, liver function
- Correct electrolyte abnormalities
- Check patient weight and height
- Confirm resuscitation equipment readily available, including Intralipid™ (20% fat emulsion) (2 x 500ml bottles) and midazolam 5mg/5ml.

2.2 Prescription

- A prescription should only be initiated and altered by:
 - an Anaesthetist,
 - an Intensive Care or Emergency Medicine Practitioner in consultation with an APS medical practitioner.
- On the **IV Analgesia Infusion Chart (in the 'Other infusions' section)**
 - **Drug: Lidocaine (lignocaine) 2 grams (20mls 10%)**
 - **Solution: 5% dextrose/ 0.9%NaCl** **Total: 500 mls**
 - **Concentration: 4mg/ml**
 - **Alternately premade bags of 0.4% lidocaine (2270mg in 567.7ml 5% dextrose are available in PACU)**
 - Rate: SEE ABOVE, charted as both mg/hr and mls/hr
 - Record any bolus given in the 'Record of administration' section of the chart
 - The charted infusion rate should be double-checked against the patient's weight (with adjustment for obesity) by two APS staff.
 - Annotate that lidocaine infusion may be switched to subcutaneous route if necessary.

2.3 Dosing

Bolus dose

Anaesthetist/Pain specialist only to administer bolus dose.

- Bolus **1.5mg/kg slow IV push over 2 – 4 minutes (if not already given)⁵.**
- Infusion 0.5 – 2 mg/kg/hour. Suggest initial 1.5mg/kg/hr⁵.
- **Duration of infusion should be guided by APS assessment of underlying pathology and ongoing need.**
 - **Suggest consideration of dose reduction beyond 24 hours (reduce by 0.5mg/kg/hr) and caution exceeding 72 hrs guided by assessment of symptoms of low level toxicity⁵.**
 - Prolonged infusion up to 14 day have been reported without toxicity⁶.
- Anaesthetist/pain specialist only to administer bolus dose.
- **Note:** If transitioning from a continuous regional/epidural infusion to lidocaine infusion, DO NOT administer a bolus dose.

Dose adjustment

- Lidocaine (lignocaine) is a high clearance drug and thus dosing depends on hepatic blood flow.
- Needs significant reduction in reduced cardiac output states.
- The dose should be based on adjusted body weight $[IBW + 0.4 \times (TBW - IBW)]$.

- An approximation is to reduce the dose as shown in the following table (bolus and infusion rate).

BMI	Lidocaine (lignocaine) dose reduction
30	10%
35	20%
40	30%

2.4 Administration

- Must have reliable IV access throughout.

2.5 Documentation

- Record that the Acute Pain Service (anaesthesia registrar on-call after hours) should be called for any concerns.
- Include to watch for perioral numbness, metallic taste, tinnitus/reduced hearing, hypotension, new onset nausea/vomiting, increasing sedation, confusion/agitation, muscular twitching.
- Usual MET criteria apply.
- The APS should be notified if: or SBP decreased by 40%, side effects, concerns.
- If required, AORC modifications will be documented by a member of the APS or anaesthesia team
- Document that the patient will be reviewed (at least) every 12 hours while the infusion is in progress.

3. Nursing Management

3.1 Pre-infusion

- Pain assessment (pain score and functional activity score)
- Record baseline vital signs:
 - blood pressure
 - pulse rate
 - respiratory rate,
 - oxygen saturation,
 - sedation score,
 - neurological state (Glasgow Coma Scale)
- Perform 12-lead ECG, lidocaine prescriber must review ECG prior to infusion commencement
- IV access flushed and accessible
- Resuscitation equipment readily accessible
 - fat emulsion 20% (intralipid): 2x 500ml bottles (located in theatre)
 - defibrillator
 - midazolam 5mg/5ml

3.2 Administration

Prepare solution

- Remove 20mls of solution from a 500ml bag of 5% dextrose / 0.9% NaCl
- Add i.e. four ampoules of 10% (500mg/5ml) lidocaine (lignocaine) i.e. 2 grams, to the remaining 480ml of 5% glucose / 0.9% NaCl
- Resulting in
 - a total volume of 500ml
 - a concentration of 4mg/ml (0.4%) Lidocaine.
- Alternately premade bags of 0.4% lidocaine (2270mg in 567.7ml) available in PACU
 - The concentration of premixed bags is 4mg/ml (0.4%) Lidocaine.

Labelling lines and route of administration

- Premix solutions (e.g. 2270mg lidocaine in 567.7ml 5% Dextrose) for bolus use or infusions that are labelled by manufacturer or hospital pharmacy DO NOT require additional labelling.
- The route of administration must be identified on all administration lines and include the date and time that the line is commenced⁵
- Labelling the medicine in a dedicated continuous infusion line: A continuous infusion line that must be retained for delivery of a single medicine must not be used for any other medicine (i.e. it is dedicated only to that single medicine, e.g. Lidocaine). The medicine contained in the line must be clearly identified alongside the route (Image 1)⁵.



Image 1: Example of line labelling – route and medicine; continuous infusion of a single medicine

Infuse via a B Braun volumetric pump. Drug Library title “Lidocaine, Acute Pain”.

DO NOT inject any other drugs through the Lidocaine infusion line.

3.3 Monitoring

- Infusion (+ loading bolus) may be commenced on the ward providing adequate staffing and monitoring available
- Pt to have 12 lead ECG prior to commencement of infusion
- The prescriber must remain present for minimum 10 minutes post bolus dose/ commencement of infusion
- Initiation of infusion requires continuous visual observation for 30 minutes with continuous SpO₂/HR and documentation of 5 minutely NIBP, HR and SpO₂ for 30 minutes.
- 12 lead ECG should be repeated and reviewed by APS medical team 20-30 minutes post commencement of infusion.
- After first 30 minutes:
 - **For 2 hours:** 30-minutely blood pressure, pulse, respiratory rate, pain scores and specific assessment for signs of local anaesthetic toxicity.
 - **Thereafter:** as per NPG # 57 post op management
 - At all times, nursing staff should be aware of and seek out symptoms of local anaesthetic toxicity.
 - Numbness around lips or face, metallic taste
 - Ringing in the ears, visual or auditory disturbance
 - Drowsiness, confusion, paranoia, dizziness, headache, muscle twitches
 - Nausea and vomiting
 - Generally feeling “unwell”
 - The APS must review at least twice daily.
 - Anaesthetists or APS medical staff only to adjust the infusion rate (in range 0.5 - 2 mg/kg/hr, noting adjustment for obesity). Rate to be increased only every 8 hours to allow steady state to be reached.

Signs of CNS or cardiac toxicity should lead to infusion cessation and supportive management according to LA toxicity protocols in line with severity of symptoms and signs.

The patient must be escorted by a nurse when away from ward for any reason, for the duration of the therapy.

If the patient is non-compliant with remaining on the ward the infusion is to be discontinued. For inadequate analgesia or any other problems concerned with therapy contact the Acute Pain Service.

4. Management of severe local anaesthetic toxicity

Reactions to lidocaine are related to high serum levels and are usually transient, however seizures and cardiac arrest may occur with severe toxicity

- With each set of observations, the patient should be assessed for:
 - Ringing in the ears
 - Numbness around lips or face
 - Metallic taste
 - Visual or auditory disturbance
 - Drowsiness, confusion, paranoia, agitation
 - Muscle twitches
 - Headache
 - Nausea and vomiting
 - Generally feeling “unwell”
 - Hypotension

Treatment of Suspected Local Anaesthetic Toxicity

Local Anaesthetic Systemic Toxicity (LAST) should be treated as per the AAGBI Safety Guidelines⁴. (Appendix 1 AAGBI)

1. **Immediately** cease lignocaine infusion if there is any evidence or suspicion of toxicity.
2. Patient should be reviewed by the treating anaesthetist or the Duty Anaesthetist if they are unavailable.
3. **Call a MET** if there is loss of consciousness, seizure, or cardiovascular collapse or as per MET criteria.
4. Follow resuscitation guidelines.
5. In the event of severe local anaesthetic toxicity Intralipid is available in Theatre – in the PACU emergency trolley and regional anaesthesia trolleys in theatre.

5. Cessation of Treatment

Cease date to be specified by Acute Pain Service on the Lidocaine Prescription, IV Analgesia Chart

Cease treatment at.....(hrs).....(date)
Date.....Sign/Print.....

The standard duration of infusion is capped at 72 hours, noting that prolonged administration may increase the likelihood of toxicity. This is at the discretion of the APS.

Procedural Information

1. Record pain score on the Lidocaine Prescription, IV Analgesia Chart.
2. Dispose of giving set and infusion bag.
3. Clean infusion pump using detergent wipes.
4. Return pump to stores.

Compliance, monitoring and evaluation

The SCGH Acute Pain Service will monitor compliance with this document. Compliance will be monitored via routine clinical incident review processes and a specific ongoing APS audit of lidocaine infusion prescribing and monitoring. The audit will prepare outcomes annually for review by the Director of APS.

Related documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010
- Work Health and Safety Act 2020

Policy

- NMHS Clinical Documentation Policy
- SCGOPHCG Infection Prevention and Management Manual -Aseptic Technique Policy.
- SCGH Patient Identification and Procedure Matching Policy
- SCGH Acute Pain Service IV lidocaine (lignocaine) for Acute Pain Policy.
- SCGOPHCG Nursing Practice Guideline IV Therapy

References

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7. Australian Commission on Safety and Quality in Health Care: National standard for user-applied labelling of injectable medicines, fluids and lines. Department of Health: DoH, WA; 2016. JBI Levels of Evidence: Level 5: Expert Opinion

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Appendix 1: AAGBI Safety Guideline – Management of Severe Local Anaesthetic Toxicity

AAGBI Safety Guideline

Management of Severe Local Anaesthetic Toxicity



1

Recognition

Signs of severe toxicity:

- Sudden alteration in mental status, severe agitation or loss of consciousness, with or without tonic-clonic convulsions
- Cardiovascular collapse: sinus bradycardia, conduction blocks, asystole and ventricular tachyarrhythmias may all occur
- Local anaesthetic (LA) toxicity may occur some time after an initial injection

2

Immediate management

- Stop injecting the LA
- Call for help
- Maintain the airway and, if necessary, secure it with a tracheal tube
- Give 100% oxygen and ensure adequate lung ventilation (hyperventilation may help by increasing plasma pH in the presence of metabolic acidosis)
- Confirm or establish intravenous access
- Control seizures: give a benzodiazepine, thiopental or propofol in small incremental doses
- Assess cardiovascular status throughout
- Consider drawing blood for analysis, but do not delay definitive treatment to do this

3

Treatment

IN CIRCULATORY ARREST

- Start cardiopulmonary resuscitation (CPR) using standard protocols
- Manage arrhythmias using the same protocols, recognising that arrhythmias may be very refractory to treatment
- Consider the use of cardiopulmonary bypass if available

GIVE INTRAVENOUS LIPID EMULSION

(following the regimen overleaf)

- Continue CPR throughout treatment with lipid emulsion
- Recovery from LA-induced cardiac arrest may take >1 h
- Propofol is not a suitable substitute for lipid emulsion
- Lidocaine should not be used as an anti-arrhythmic therapy

WITHOUT CIRCULATORY ARREST

Use conventional therapies to treat:

- hypotension,
- bradycardia,
- tachyarrhythmia

CONSIDER INTRAVENOUS LIPID EMULSION

(following the regimen overleaf)

- Propofol is not a suitable substitute for lipid emulsion
- Lidocaine should not be used as an anti-arrhythmic therapy

4

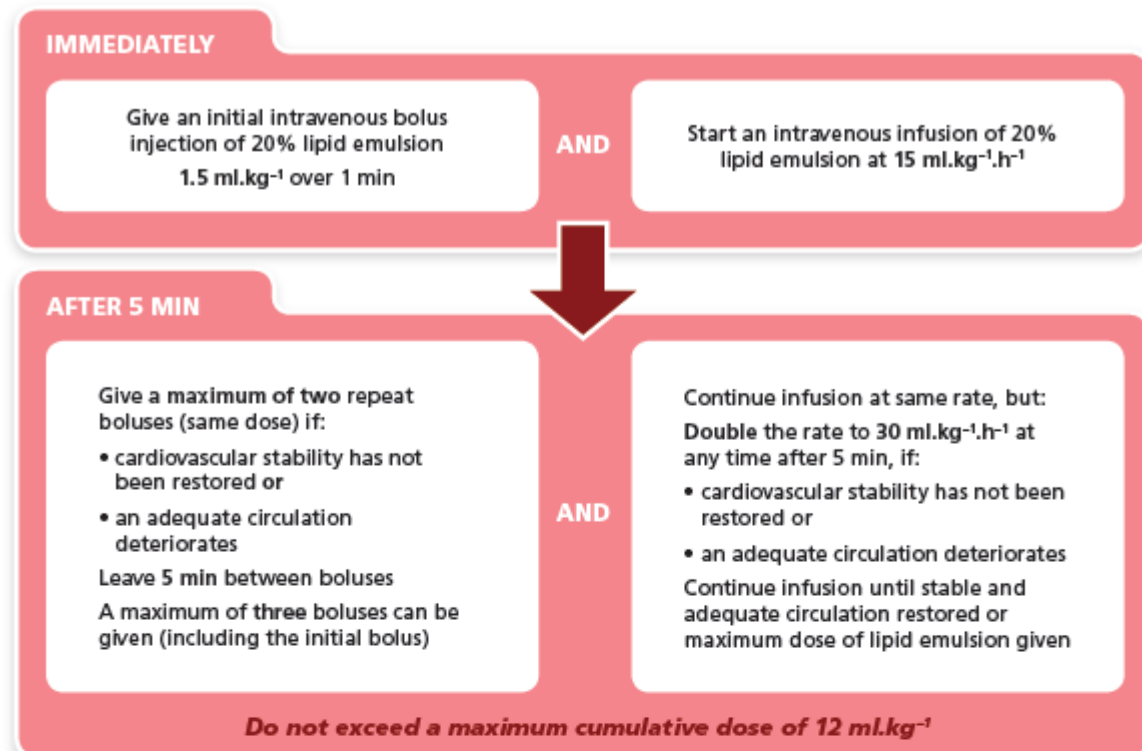
Follow-up

- Arrange safe transfer to a clinical area with appropriate equipment and suitable staff until sustained recovery is achieved
 - Exclude pancreatitis by regular clinical review, including daily amylase or lipase assays for two days
 - Report cases as follows:
 - in the United Kingdom to the National Patient Safety Agency (via www.npsa.nhs.uk)
 - in the Republic of Ireland to the Irish Medicines Board (via www.imb.ie)
- If Lipid has been given, please also report its use to the international registry at www.lipidregistry.org. Details may also be posted at www.lipidrescue.org

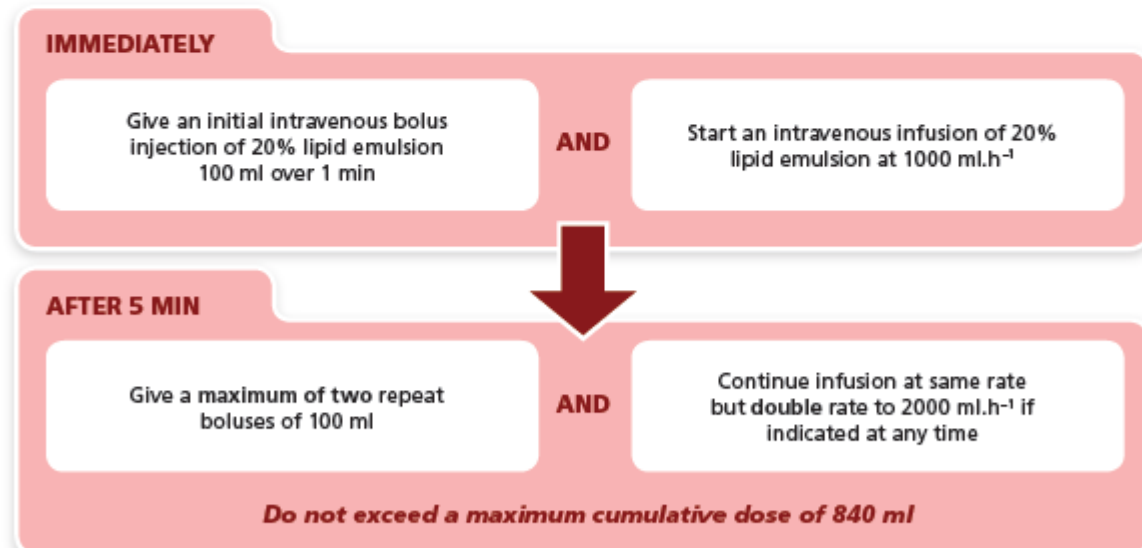
Your nearest bag of Lipid Emulsion is kept

This guideline is not a standard of medical care. The ultimate judgement with regard to a particular clinical procedure or treatment plan must be made by the clinician in the light of the clinical data presented and the diagnostic and treatment options available.

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An approximate dose regimen for a 70-kg patient would be as follows:



This AAGBI Safety Guideline was produced by a Working Party that comprised:
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